B3.1 Gas exchange in plants

NAME	DATE	TOTAL
		_____ / 35

About 35 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 **STATE** the function of the waxy cuticle. [B3.1.7] [1]

2 **DISTINGUISH** between a stoma and a guard cell. [B3.1.7] [2]

3 **DEFINE** transpiration. [B3.1.9] [1]

4 **STATE** the effect on the rate of transpiration of an increase in humidity and an increase in wind. [B3.1.9] [2]

5 **DEFINE** stomatal density. [B3.1.10] [1]

Section B: Understanding

10 MARKS

6 **EXPLAIN** how the spongy mesophyll and air spaces help gas exchange. [B3.1.7] [2]

7 **EXPLAIN** why gas exchange in a leaf leads to water loss. [B3.1.9] [3]

8 DRAW a labelled plan diagram of a transverse section of a dicot leaf, showing four tissues. [B3.1.8] [4]

9 STATE why individual cells should not be drawn in a plan diagram. [B3.1.8] [1]

Section C: Working with data

18 MARKS

All data in this section are invented for practice.

10 [B3.1.9]

A student used a potometer. The air bubble moved along a capillary tube with these results.

CONDITION	DISTANCE MOVED IN 5 MINUTES / MM	RATE / MM MIN^{-1}
STILL AIR	20	
MOVING AIR (FAN)	65	
PLASTIC BAG OVER SHOOT	5	

a CALCULATE the three missing rates. [2]

b EXPLAIN the result with moving air. [2]

c EXPLAIN the result with the plastic bag. [2]

d STATE two variables that should be kept constant. [2]

e OUTLINE why a potometer does not directly measure transpiration. [2]

11 [B3.1.10]

Leaf casts were made from the lower surface of leaves from two plant species. The field of view had a diameter of 0.50 mm.

SPECIES	FIELD 1	FIELD 2	FIELD 3
X (WOODLAND)	12	10	14
Y (FULL SUN)	30	26	28

a CALCULATE the area of the field of view. [1]

b CALCULATE the stomatal density of species Y. [2]

c SUGGEST one advantage to species Y of its higher density. [1]

d SUGGEST two ways to improve the reliability of this investigation. [2]

e SUGGEST a disadvantage of a high stomatal density in a dry habitat. [2]